EXCEL MASTER INDEX

Basic to Advanced | English & Roman Hindi

EXCEL LOOKUP AND REFERENCE FORMULAS

VLOOKUP, HLOOKUP, XLOOKUP, LOOKUP, INDEX, MATCH, XMATCH, INDEX+MATCH, OFFSET, INDIRECT, CHOOSE, TRANSPOSE | Bilingual Guide

Created & Developed by

UTKARSH BAJPAI

utkarshbajpai068@gmail.com

Premium Bilingual Resource | For Students, MIS Executives, Data Analysts & Office Professionals

TABLE OF CONTENTS

1. VLOOKUP
2. HLOOKUP
3. XLOOKUP
4. LOOKUP
5. INDEX
6. MATCH
7. XMATCH
8. INDEX + MATCH
9. INDEX + XMATCH
10. OFFSET
11. INDIRECT
12. CHOOSE
13. TRANSPOSE
14. ROW
15. ROWS
16. COLUMN
17. COLUMNS
18. HYPERLINK

UTKARSH BAJPAI

CHAPTER 1: VLOOKUP

PURPOSE - ENGLISH

Searches the first column of a table and returns a value from another column.

PURPOSE - ROMAN HINDI

Table ka pehla column search karta hai aur dusri column ki value deta hai.

SYNTAX

```
=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
```

SYNTAX EXPLANATION

VLOOKUP looks up data vertically in the first column.

Roman Hindi: VLOOKUP pehle column me data dhoondta hai.

SAMPLE DATASET

Cells: A2:C5 | Values: 101,Rahul,Delhi
102,Priya,Mumbai
103,Arjun,Delhi

ACTUAL FORMULA

```
=VLOOKUP(A2, A2:C5, 2, FALSE)
```

EXPECTED OUTPUT

"Rahul"

STEP-BY-STEP - ENGLISH

- Have lookup value and table.
- Use =VLOOKUP(A2, B2:C5, 2, FALSE).
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- Lookup value aur table dena.
- =VLOOKUP(A2, B2:C5, 2, FALSE) likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

Used to find employee or product details.

Roman Hindi: Employee ya product ki details dhoone ke liye use hota hai.

COMMON MISTAKES

- Lookup column must be first.

UTKARSH BAJPAI

IMPORTANT TIPS

- Use FALSE for exact match.

EXCEL VERSION COMPATIBILITY

Excel 2007+

CHAPTER 2: HLOOKUP

PURPOSE - ENGLISH

Searches the first row of a table and returns a value.

PURPOSE - ROMAN HINDI

Table ka pehla row search karta hai aur value deta hai.

SYNTAX

```
=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])
```

SYNTAX EXPLANATION

HLOOKUP looks up horizontally.

Roman Hindi: HLOOKUP horizontal tarike se dhoondta hai.

SAMPLE DATASET

Cells: A1:C2 | Values: ID,Name,City
101,Rahul,Delhi

ACTUAL FORMULA

```
=HLOOKUP("Name", A1:C2, 2, FALSE)
```

EXPECTED OUTPUT

"Rahul"

STEP-BY-STEP - ENGLISH

- Look for column header.
- Use =HLOOKUP("Name", A1:C2, 2, FALSE).
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- Column header pe dhoondo.
- =HLOOKUP("Name", A1:C2, 2, FALSE) likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

UTKARSH BAJPAI

Used for header based reports.

Roman Hindi: Header based reports ke liye use hota hai.

COMMON MISTAKES

- Row index must be right.

IMPORTANT TIPS

- Use FALSE for exact match.

EXCEL VERSION COMPATIBILITY

Excel 2007+

CHAPTER 3: XLOOKUP

UTKARSH BAJPAI

PURPOSE - ENGLISH

Searches any column and returns a value; modern replacement of VLOOKUP.

PURPOSE - ROMAN HINDI

Kisi bhi column me dhoondta hai aur value deta hai; VLOOKUP ka modern replacement.

SYNTAX

```
=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found])
```

SYNTAX EXPLANATION

XLOOKUP searches flexibly and handles errors.

Roman Hindi: XLOOKUP flexibly dhoonda hai aur errors handle karta hai.

SAMPLE DATASET

Cells: B2:C5 | Values: Rahul,Delhi
Priya,Mumbai
Arjun,Delhi

ACTUAL FORMULA

```
=XLOOKUP("Rahul", B2:B5, C2:C5, "Not Found")
```

EXPECTED OUTPUT

"Delhi"

STEP-BY-STEP - ENGLISH

- Add lookup, return array and fallback.
- Use =XLOOKUP("Rahul", B2:B5, C2:C5, "Not Found").
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- Lookup, return array aur fallback dein.
- =XLOOKUP("Rahul", B2:B5, C2:C5, "Not Found") likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

Used for employee lookup in modern offices.

Roman Hindi: Modern offices me employee lookup ke liye use hota hai.

COMMON MISTAKES

- Requires Microsoft 365 or Excel 2021.

IMPORTANT TIPS

- if_not_found handles missing values.

EXCEL VERSION COMPATIBILITY

Microsoft 365 / Excel 2021 (Microsoft 365 / newer version only)

CHAPTER 4: LOOKUP

PURPOSE - ENGLISH

Performs an approximate or exact lookup.

PURPOSE - ROMAN HINDI

Approximate ya exact lookup karta hai.

SYNTAX

```
=LOOKUP(lookup_value, lookup_array, result_array)
```

SYNTAX EXPLANATION

LOOKUP does a lookup from first row or column.

Roman Hindi: LOOKUP pehla row ya column se lookup karta hai.

SAMPLE DATASET

Cells: A1 | Values: 2

ACTUAL FORMULA

```
=LOOKUP(2, {1,2,3,4}, {"Basic","Intermediate","Advanced","Expert"})
```

EXPECTED OUTPUT

"Intermediate"

UTKARSH BAJPAI

STEP-BY-STEP - ENGLISH

- Add lookup value and arrays.
- Use =LOOKUP(2, {1,2,3,4}, {"Basic",...}).
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- Lookup value aur arrays dein.
- =LOOKUP(2, {1,2,3,4}, {"Basic","Intermediate","Advanced","Expert"}) likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

Used for simple category mapping.

Roman Hindi: Simple category mapping ke liye use hota hai.

COMMON MISTAKES

- Approximate match needs sorted data.

IMPORTANT TIPS

- Exact match use karein.

EXCEL VERSION COMPATIBILITY

Excel 2007+

CHAPTER 5: INDEX

PURPOSE - ENGLISH

Returns a value from a table using row and column numbers.

PURPOSE - ROMAN HINDI

Row aur column number se value nikalta hai.

SYNTAX

```
=INDEX(array, row_num, [column_num])
```

SYNTAX EXPLANATION

INDEX fetches a cell by position.

Roman Hindi: INDEX position ke mutabik cell deta hai.

SAMPLE DATASET

Cells: A1:C4 | Values: ID,Name,City
101,Rahul,Delhi
102,Priya,Mumbai
103,Arjun,Delhi

UTKARSH BAJPAI

ACTUAL FORMULA

```
=INDEX(B2:C4, 2, 2)
```

EXPECTED OUTPUT

"Mumbai"

STEP-BY-STEP - ENGLISH

- Add array, row and column.
- Use =INDEX(B2:C3, 2, 2).
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- Array, row aur column dein.
- =INDEX(B2:C3, 2, 2) likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

Used for dynamic row lookups.

Roman Hindi: Dynamic row lookups ke liye use hota hai.

COMMON MISTAKES

- Wrong row or column number.

IMPORTANT TIPS

- Combine with MATCH for true lookups.

EXCEL VERSION COMPATIBILITY

Excel 2007+

CHAPTER 6: MATCH

PURPOSE - ENGLISH

Finds the position of a value in a range.

PURPOSE - ROMAN HINDI

Kisi range me value ki position dhoondta hai.

SYNTAX

```
=MATCH(lookup_value, lookup_array, [match_type])
```

SYNTAX EXPLANATION

MATCH returns position of a value.

UTKARSH BAJPAI

Roman Hindi: MATCH value ki position deta hai.

SAMPLE DATASET

Cells: A2:A5 | Values: 101
102
103
104

ACTUAL FORMULA

```
=MATCH(102, A2:A5, 0)
```

EXPECTED OUTPUT

2

STEP-BY-STEP - ENGLISH

- Add lookup value and range.
- Use =MATCH(102, A2:A5, 0).
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- Lookup value aur range dein.
- =MATCH(102, A2:A5, 0) likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

Used to find rows for INDEX.

Roman Hindi: INDEX ke liye rows dhoone ke liye use hota hai.

COMMON MISTAKES

- match_type 0 for exact.

IMPORTANT TIPS

- Perfect for INDEX + MATCH.

EXCEL VERSION COMPATIBILITY

Excel 2007+

CHAPTER 7: XMATCH

PURPOSE - ENGLISH

Finds position of a value with optional exact/approximate.

PURPOSE - ROMAN HINDI

Value ki position dhoondta hai optional exact/approximate ke saath.

UTKARSH BAJPAI

SYNTAX

```
=XMATCH(lookup_value, lookup_array, [match_mode], [search_mode])
```

SYNTAX EXPLANATION

XMATCH is the extended MATCH.

Roman Hindi: XMATCH MATCH ka extended version hai.

SAMPLE DATASET

Cells: A2:A5 | Values: 101
102
103
104

ACTUAL FORMULA

```
=XMATCH(102, A2:A5, 0)
```

EXPECTED OUTPUT

2

STEP-BY-STEP - ENGLISH

- Add lookup value and array.
- Use =XMATCH(102, A2:A5, 0).
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- Lookup value aur array dein.
- =XMATCH(102, A2:A5, 0) likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

Used for flexible lookups.

Roman Hindi: Flexible lookups ke liye use hota hai.

COMMON MISTAKES

- Requires Microsoft 365 or Excel 2021.

IMPORTANT TIPS

- Handles errors better.

EXCEL VERSION COMPATIBILITY

Microsoft 365 / Excel 2021 (Microsoft 365 / newer version only)

CHAPTER 8: INDEX + MATCH

UTKARSH BAJPAI

PURPOSE - ENGLISH

Combines INDEX and MATCH for flexible lookups.

PURPOSE - ROMAN HINDI

Flexible lookups ke liye INDEX aur MATCH ka combination.

SYNTAX

```
=INDEX(return_array, MATCH(lookup_value, lookup_array, 0))
```

SYNTAX EXPLANATION

INDEX + MATCH is a classic flexible lookup.

Roman Hindi: INDEX + MATCH classic flexible lookup hai.

SAMPLE DATASET

Cells: A2:B5 | Values: 101,Rahul
102,Priya
103,Arjun
104,Meera

ACTUAL FORMULA

```
=INDEX(B2:B5, MATCH(103, A2:A5, 0))
```

EXPECTED OUTPUT

"Arjun"

STEP-BY-STEP - ENGLISH

- Use MATCH to find position.
- Wrap with INDEX.
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- MATCH se position dhojo.
- INDEX se is tarah use karo.
- Enter dabao.

REAL OFFICE EXAMPLE

Used before XLOOKUP for complex lookups.

Roman Hindi: Complex lookups ke liye XLOOKUP se pehle use hota tha.

COMMON MISTAKES

- Forgetting the 0 for exact match.

IMPORTANT TIPS

- Works in all Excel versions.

UTKARSH BAJPAI

EXCEL VERSION COMPATIBILITY

Excel 2007+

CHAPTER 9: INDEX + XMATCH

PURPOSE - ENGLISH

Combines INDEX and XMATCH for advanced lookups.

PURPOSE - ROMAN HINDI

Advanced lookups ke liye INDEX aur XMATCH ka combination.

SYNTAX

```
=INDEX(return_array, XMATCH(lookup_value, lookup_array))
```

SYNTAX EXPLANATION

INDEX + XMATCH is modern flexible lookup.

Roman Hindi: INDEX + XMATCH modern flexible lookup hai.

SAMPLE DATASET

Cells: A2:B5 | Values: 101,Rahul
102,Priya
103,Arjun
104,Meera

ACTUAL FORMULA

```
=INDEX(B2:B5, XMATCH(103, A2:A5))
```

EXPECTED OUTPUT

"Arjun"

STEP-BY-STEP - ENGLISH

- Use XMATCH to find position.
- Wrap with INDEX.
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- XMATCH se position dhojo.
- INDEX se is tarah use karo.
- Enter dabao.

REAL OFFICE EXAMPLE

Used for large data sets.

UTKARSH BAJPAI

Roman Hindi: Bade data sets ke liye use hota hai.

COMMON MISTAKES

- Requires Microsoft 365 or Excel 2021.

IMPORTANT TIPS

- More flexible error handling.

EXCEL VERSION COMPATIBILITY

Microsoft 365 / Excel 2021 (Microsoft 365 / newer version only)

CHAPTER 10: OFFSET

PURPOSE - ENGLISH

Returns a reference offset from a starting cell.

PURPOSE - ROMAN HINDI

Starting cell se offset par reference deta hai.

SYNTAX

```
=OFFSET(reference, rows, cols, [height, width])
```

SYNTAX EXPLANATION

OFFSET moves from a base cell.

Roman Hindi: OFFSET base cell se move karta hai.

SAMPLE DATASET

Cells: A1:A3 | Values: 5
10
15

ACTUAL FORMULA

```
=OFFSET(A1, 2, 0)
```

EXPECTED OUTPUT

15

STEP-BY-STEP - ENGLISH

- Add row and column offset.
- Use =OFFSET(A1, 2, 0).
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

UTKARSH BAJPAI

- Row aur column offset dein.
- =OFFSET(A1, 2, 0) likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

Used for dynamic ranges.

Roman Hindi: Dynamic ranges ke liye use hota hai.

COMMON MISTAKES

- Volatile function.

IMPORTANT TIPS

- Not allowed in tables.

EXCEL VERSION COMPATIBILITY

Excel 2007+

CHAPTER 11: INDIRECT

PURPOSE - ENGLISH

Returns a reference from text.

PURPOSE - ROMAN HINDI

Text se reference nikalta hai.

SYNTAX

```
=INDIRECT(ref_text, [is_A1])
```

SYNTAX EXPLANATION

INDIRECT builds a reference from text.

Roman Hindi: INDIRECT text se reference banata hai.

SAMPLE DATASET

Cells: A1 | Values: 5

ACTUAL FORMULA

```
=INDIRECT("A1")
```

EXPECTED OUTPUT

5

STEP-BY-STEP - ENGLISH

- Pass the reference text.

UTKARSH BAJPAI

- Use =INDIRECT("A1").
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- Reference text dein.
- =INDIRECT("A1") likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

Used to build references from names.

Roman Hindi: Names se references banane ke liye use hota hai.

COMMON MISTAKES

- Text must be valid.

IMPORTANT TIPS

- Not volatile friendly in old Excel.

EXCEL VERSION COMPATIBILITY

Excel 2007+

CHAPTER 12: CHOOSE

PURPOSE - ENGLISH

Returns a value from a list based on index.

PURPOSE - ROMAN HINDI

Index ke mutabik list se value deta hai.

SYNTAX

```
=CHOOSE(index_num, value1, [value2], ...)
```

SYNTAX EXPLANATION

CHOOSE picks a value.

Roman Hindi: CHOOSE koi value select karta hai.

SAMPLE DATASET

Cells: A1 | Values: 2

ACTUAL FORMULA

```
=CHOOSE(2, "A", "B", "C")
```

EXPECTED OUTPUT

UTKARSH BAJPAI

"B"

STEP-BY-STEP - ENGLISH

- Add index and values.
- Use =CHOOSE(2, "A", "B", "C").
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- Index aur values dein.
- =CHOOSE(2, "A", "B", "C") likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

Used for simple conditional picks.

Roman Hindi: Simple conditional picks ke liye use hota hai.

COMMON MISTAKES

- Index out of range.

IMPORTANT TIPS

- Good for small logic lists.

EXCEL VERSION COMPATIBILITY

Excel 2007+

CHAPTER 13: TRANSPOSE

PURPOSE - ENGLISH

Returns the transpose of a range.

PURPOSE - ROMAN HINDI

Range ka transpose deta hai.

SYNTAX

```
=TRANSPOSE(array)
```

SYNTAX EXPLANATION

TRANSPOSE flips rows and columns.

Roman Hindi: TRANSPOSE rows aur columns ko phir se le leta hai.

SAMPLE DATASET

Cells: A1:B2 | Values: 1,2
3,4

UTKARSH BAJPAI

ACTUAL FORMULA

```
=TRANSPOSE(A1:B2)
```

EXPECTED OUTPUT

{1,3;2,4}

STEP-BY-STEP - ENGLISH

- Select the range.
- Use =TRANSPOSE(A1:B2).
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- Range select karo.
- =TRANSPOSE(A1:B2) likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

Used for matrix math.

Roman Hindi: Matrix math ke liye use hota hai.

COMMON MISTAKES

- Spills to new cells.

IMPORTANT TIPS

- Needs enough empty cells.

EXCEL VERSION COMPATIBILITY

Excel 2007+

CHAPTER 14: ROW

PURPOSE - ENGLISH

Returns the row number of a cell.

PURPOSE - ROMAN HINDI

Cell ki row number deta hai.

SYNTAX

```
=ROW([reference])
```

SYNTAX EXPLANATION

ROW returns row number.

UTKARSH BAJPAI

Roman Hindi: ROW row number deta hai.

SAMPLE DATASET

Cells: A1 | Values: 5

ACTUAL FORMULA

```
=ROW(A1)
```

EXPECTED OUTPUT

1

STEP-BY-STEP - ENGLISH

- Pass the cell.
- Use =ROW(A1).
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- Cell pass karo.
- =ROW(A1) likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

Used for row based logic.

Roman Hindi: Row based logic ke liye use hota hai.

COMMON MISTAKES

- None.

IMPORTANT TIPS

- Can return multiple rows.

EXCEL VERSION COMPATIBILITY

Excel 2007+

UTKARSH BAJPAI

CHAPTER 15: ROWS

PURPOSE - ENGLISH

Returns the number of rows in a range.

PURPOSE - ROMAN HINDI

Range me kitne rows hain unki ginti karta hai.

SYNTAX

```
=ROWS(array)
```

SYNTAX EXPLANATION

ROWS counts rows.

Roman Hindi: ROWS rows count karta hai.

SAMPLE DATASET

Cells: A1:A5 | Values: 5, 10, 15, 20, 25

ACTUAL FORMULA

```
=ROWS(A1:A5)
```

EXPECTED OUTPUT

5

STEP-BY-STEP - ENGLISH

- Pass the range.
- Use =ROWS(A1:A5).
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- Range pass karo.
- =ROWS(A1:A5) likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

Used for dynamic dashboards.

Roman Hindi: Dynamic dashboards ke liye use hota hai.

COMMON MISTAKES

- None.

IMPORTANT TIPS

- Good with INDEX.

EXCEL VERSION COMPATIBILITY

Excel 2007+

CHAPTER 16: COLUMN

PURPOSE - ENGLISH

Returns the column number of a cell.

PURPOSE - ROMAN HINDI

Cell ki column number deta hai.

UTKARSH BAJPAI

SYNTAX

```
=COLUMN([reference])
```

SYNTAX EXPLANATION

COLUMN returns column number.

Roman Hindi: COLUMN column number deta hai.

SAMPLE DATASET

Cells: A1 | Values: 5

ACTUAL FORMULA

```
=COLUMN(B2)
```

EXPECTED OUTPUT

2

STEP-BY-STEP - ENGLISH

- Pass the cell.
- Use =COLUMN(B2).
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- Cell pass karo.
- =COLUMN(B2) likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

Used for column based logic.

Roman Hindi: Column based logic ke liye use hota hai.

COMMON MISTAKES

- None.

IMPORTANT TIPS

- Returns first column as 1.

EXCEL VERSION COMPATIBILITY

Excel 2007+

CHAPTER 17: COLUMNS

PURPOSE - ENGLISH

UTKARSH BAJPAI

Returns the number of columns in a range.

PURPOSE - ROMAN HINDI

Range me kitne columns hain unki ginti karta hai.

SYNTAX

```
=COLUMNS(array)
```

SYNTAX EXPLANATION

COLUMNS counts columns.

Roman Hindi: COLUMNS columns count karta hai.

SAMPLE DATASET

Cells: A1:B2 | Values: 5, 10

ACTUAL FORMULA

```
=COLUMNS(A1:B2)
```

EXPECTED OUTPUT

2

STEP-BY-STEP - ENGLISH

- Pass the range.
- Use =COLUMNS(A1:B2).
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- Range pass karo.
- =COLUMNS(A1:B2) likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

Used for matrix operations.

Roman Hindi: Matrix operations ke liye use hota hai.

COMMON MISTAKES

- None.

IMPORTANT TIPS

- Useful with TRANSPOSE.

EXCEL VERSION COMPATIBILITY

Excel 2007+

UTKARSH BAJPAI

CHAPTER 18: HYPERLINK

PURPOSE - ENGLISH

Creates a clickable hyperlink in a cell.

PURPOSE - ROMAN HINDI

Cell me clickable hyperlink banata hai.

SYNTAX

```
=HYPERLINK(url, friendly_name)
```

SYNTAX EXPLANATION

HYPERLINK makes a link.

Roman Hindi: HYPERLINK link banata hai.

SAMPLE DATASET

Cells: A1 | Values: https://example.com

ACTUAL FORMULA

```
=HYPERLINK("https://example.com", "Click Here")
```

EXPECTED OUTPUT

Click Here

STEP-BY-STEP - ENGLISH

- Add URL and friendly text.
- Use =HYPERLINK("https://example.com", "Click Here").
- Press Enter.

STEP-BY-STEP - ROMAN HINDI

- URL aur friendly text dein.
- =HYPERLINK("https://example.com", "Click Here") likhna.
- Enter dabao.

REAL OFFICE EXAMPLE

Used for report portal links.

Roman Hindi: Report portal links ke liye use hota hai.

COMMON MISTAKES

- URL must include protocol.

IMPORTANT TIPS

- Opens in browser.

UTKARSH BAJPAI

EXCEL VERSION COMPATIBILITY

Excel 2007+

UTKARSH BAJPAI